FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VIII

JEE (Main)-2020

TEST DATE: 27-08-2020

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. B

Sol. $a = \frac{mg\sin\theta + f_k}{m} = \frac{mg\sin\theta + mg\sin\theta}{m}$

$a = 2g\sin\theta$

$\tau_{CM} = I_{CM}\alpha$

$f_k R = \frac{2}{5}mR^2\alpha$

$\alpha = \frac{5g\sin\theta}{2R}$

When pure rolling motion starts

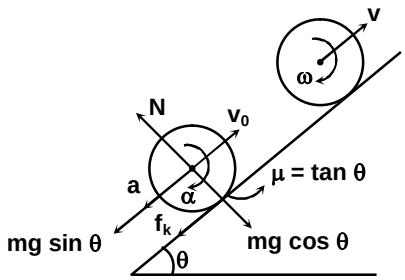
$v = \omega R$

$v_0 - at = \alpha tR$

$v_0 - 2g\sin\theta t = \frac{5g\sin\theta t}{2}$

$v_0 = \frac{9g\sin\theta t}{2}$

$t = \frac{2v_0}{9g\sin\theta}$



2. A

Sol. $\int\tau\cdot dt = I\omega$

$\int 2qE\ell dt = I\omega$

$\int 2q\ell\frac{R^2}{2\ell}\frac{dB}{dt}\cdot dt = \left(\frac{4m\ell^2}{12} + m\ell^2 + m\ell^2\right)\omega$

$$qR^2B = \frac{7m\ell^2}{3}\omega$$

$$\omega = \left(\frac{3qBR^2}{7m\ell^2}\right)$$

3. A

Sol. $\phi = \frac{\sqrt{3}}{4}\ell^2 B\sin\omega t$

$$|\varepsilon| = \frac{\sqrt{3}}{4}\ell^2\omega B\cos\omega t$$

$$I = \frac{\sqrt{3}\ell^2\omega B}{4R}\cos\omega t$$

$$P_{av} = \frac{\int_0^{2\pi/\omega} I^2 R dt}{\int_0^{2\pi/\omega} dt} = \left(\frac{\sqrt{3}\ell^2 B\omega}{4R}\right)^2 R\frac{\int_0^{\frac{2\pi}{\omega}} \cos^2\omega t dt}{\int_0^{2\pi/\omega} dt}$$

$$P_{av} = \left(\frac{\sqrt{3}\ell^2 B\omega}{4R}\right)^2 R\frac{1}{2} = \frac{3B^2\omega^2\ell^4}{32R}$$

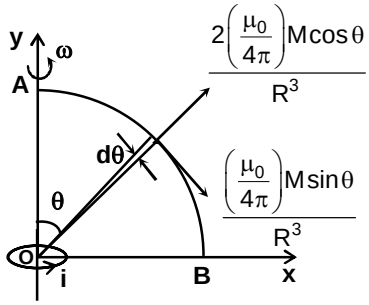
4. B

Sol. $v = R\sin\theta\cdot\omega$

$$\int_0^{\pi/2} \frac{2\mu_0}{4\pi R^3} M\cos\theta\cdot R\sin\theta\cdot\omega R d\theta$$

$$= \frac{\mu_0 M\omega}{2\pi\times 2R}\int_0^{\pi/2}\sin 2\theta\cdot d\theta = \frac{\mu_0 M\omega}{4\pi R}\left(-\frac{\cos 2\theta}{2}\right)_0^{\pi/2}$$

$$\frac{\mu_0 M\omega}{4\pi R}\left(\frac{1}{2}+\frac{1}{2}\right) = \frac{\mu_0 M\omega}{4\pi R} = \frac{\mu_0\omega i\pi a^2}{4\pi R} = \frac{\mu_0 a^2\omega i}{4R}$$



5. C

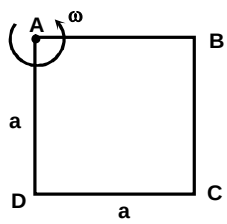
Sol. $F = M\left|\frac{dB}{dr}\right| = \frac{\mu_0}{4\pi}\frac{6M^2}{d^4} = I\pi R^2\left(\frac{3\mu_0 IR^2}{2d^4}\right)$

6. B

Sol. Using conservation of angular momentum about the vertical axis passing through vertex 'A'.

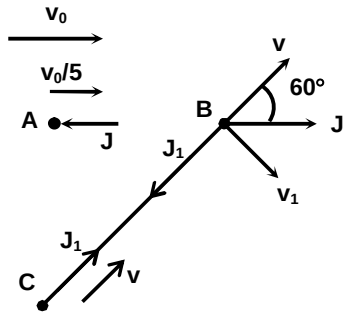
$$mv\frac{a}{2} + mv\frac{a}{2} = \frac{2}{3}ma^2\omega$$

$$\Rightarrow mva = \frac{2}{3}ma^2\omega \quad \Rightarrow \omega = \frac{3v}{2a}$$



7. D

Sol. $J = m\left(v_0 - \frac{v_0}{5}\right)$



$J = \frac{4mv_0}{5}$

$J\cos 60° = 2mv$

$\frac{2mv_0}{5} = 2mv$

$v = \frac{v_0}{5}$

Impulse due to string on the ball 'C' during collision is

$J_1 = mv$

$\Rightarrow J_1 = \frac{mv_0}{5}$

8. B

Sol. The wavelength of sound received by the observer is

$\lambda = \frac{4\left(v - \frac{v}{5}\right)}{f} = \frac{16v}{5f}$

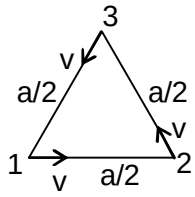
9. B

Sol. $F = \int_{h/2}^{h} (P_0 + \rho gy)2Rdy = \left(P_0 + \rho g\frac{3h}{4}\right)Rh$

10. D

Sol. Acceleration of each particle

$a_0 = v\omega = v\frac{v\sin 60°}{a/2} = \frac{\sqrt{3}v^2}{a}$



11. B

Sol. $80 = 10\log\frac{I_1}{I_0}$ …(i)

$40 = 10\log\frac{I_2}{I_0}$ ….(ii)

From (i) and (ii)

$40 = 10\log\frac{I_1}{I_2}$

$4 = \log\left(\frac{r_2}{r_1}\right)^2$

$\frac{r_2}{r_1} = 10^2$

$r_2 = 10 \times 10^2$ cm = 10 m

12. A

Sol. mobility will remain same.

13. D

Sol. Frequency of the sonometer wire, $f = \frac{2}{2\ell}\sqrt{\frac{F}{\mu}} = \frac{1}{0.17}\sqrt{\frac{289}{1}} = 100$ Hz

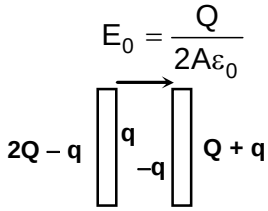
Frequency of the tuning fork is $f_0 = f + 5 = 105$ Hz

14. C

Sol. $\left(\frac{2Q-q}{2\varepsilon_0 A}\right)+\left(\frac{Q}{2\varepsilon_0 A}\right)=\left(\frac{Q+q}{2\varepsilon_0 A}\right)$

$(2Q - q) + Q = Q + q$

$q = Q$



15. B

Sol. $I = \frac{I_0}{2}\cos^2 37° \times \cos^2\theta = \frac{I_0 8}{100}$

$\frac{I_0}{2}\times\frac{16}{25}\times\cos^2\theta = \frac{I_0\times 4}{50}$

$\cos^2\theta = \frac{1}{4}$

$\cos\theta = \frac{1}{2}$

$\theta = 60°$

16. B

Sol. $\sin 60° = \mu \sin r$

$\Rightarrow \frac{\sqrt{3}}{2} = \sqrt{3}\sin r$

$r = 30°$

The net deviation produced in the ray

$\delta = 30° + 100° + 30° = 160°$ clockwise

17. A

Sol. L.S.B. = (1000 – 5)kHz = 995 kHz

U.S.B. = (1000 + 5) kHz = 1005 kHz

18. C

Sol. $m = \rho\pi r^2\ell$

$100\times\frac{\Delta m}{m} = \left(\frac{\Delta\rho}{\rho}+2\frac{\Delta r}{r}+\frac{\Delta\ell}{\ell}\right)\times 100 = \left(\frac{0.004}{0.4}+2\times\frac{0.006}{0.6}+\frac{0.05}{5}\right)\times 100$

Maximum % error in the mass = $(0.01 + 2 \times 0.01 + 0.01) \times 100 = 4\%$

19. C

Sol. $(\mu_1 - \mu_2)t = 3\lambda$

$t = \frac{3\lambda}{(\mu_1-\mu_2)} = \frac{3\times 5\times 10^{-7}}{(1.8-1.5)} = 5\ \mu m$

20. A

Sol. The equation of process is $P = \alpha\sqrt{V}$

$PV^{-1/2} = \text{constant}$

This is a polytropic process with a polytropic constant, x = –(1/2)

∴ Molar heat capacity of the gas

$$C = C_V + \frac{R}{(1-x)} = \frac{5R}{2} + \frac{R}{\left(1+\frac{1}{2}\right)} = \frac{5R}{2} + \frac{2R}{3} = \frac{19R}{6}$$

SECTION – B

21. 7

Sol. $\frac{4}{3}\sin 37° = 1 \times \sin r$

r = 53°

So, x = 4 tan 37° + 3 cot 37° = 7 m

22. 2

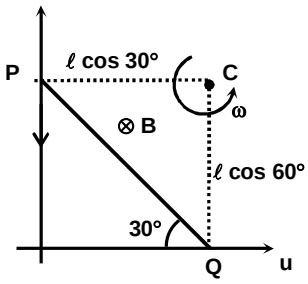
Sol. Angular velocity of the rod is

$$\omega = \frac{u}{\ell \cos 60°} = \frac{2u}{\ell}$$

Potential difference between P and Q

$$= \frac{1}{2}B\omega\ell^2\left(\cos^2 30° - \cos^2 60°\right) = \frac{Bu\ell}{2}$$

∴ n = 2



SECTION – C

23. 00003.25

Sol. Pitch of the screw gauge, P = 1mm

Least count, L.C. = $\frac{P}{N} = \frac{1}{100} = 0.01\text{mm}$

Diameter of the wire, d = 3 mm + 25 × L.C. = 3mm + 25 × 0.01 mm

∴ d = 3.25 mm

24. 00020.00

Sol. $I = \frac{6}{300} = 0.02$ ampere = 20 milliampere

25. 00028.20

Sol. $$T = 2\pi\sqrt{\frac{\frac{mR^2}{2} + mx^2}{mgx}}$$

T will be minimum for $x = \frac{R}{\sqrt{2}} = \frac{40}{\sqrt{2}} = 20\sqrt{2}$ = 28.2 cm

Chemistry

PART – II

SECTION – A

26. D

Sol. 2.76×10^{-3}

Subtraction of second equation from first will give the required equation. Hence

$\Delta G = -41.0-(-55.6)$

$=14.6 \text{ kJ} = -2.303RT \log \text{K}$

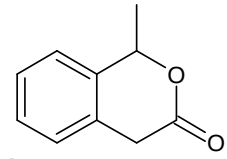
$\Rightarrow \log_{10} \text{K} = -\dfrac{14.6\times1000}{8.3\times298\times2.303}$

$=-2.56$

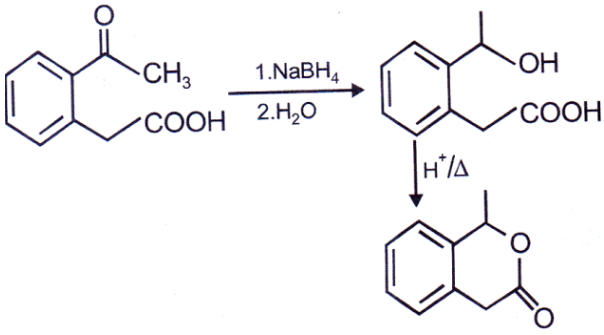
$\Rightarrow \text{K}=2.7\times10^{-3}$

27. C

Sol.



I.



28. D

Sol. $Z = 4\times0.8 = 3.2$

$\dfrac{a}{2}=100$

$a = 200 \text{ pm}$

$d=\dfrac{Z\times M}{N_A\times a^3}=\dfrac{3.2\times60}{6\times10^{23}\times\left(2\times10^{-8}\right)^3}$

29. B

Sol. A will be faster both in S_N1 and S_N2 reaction

Allylic halide gives both S_N1 and S_N2 reaction

30. A

Sol. I, II

31. C

Sol. At 298 K,

$Zn(s) + 2H^+(aq) \rightarrow Zn^{2+}(0.1M) + H_2(g); E° = 0.76V$

EMF = 0.199 V

From Nernst equation,

$$E_{cell} = E^0_{Cell} + \frac{0.0591}{n}\log\frac{\left[H^+\right]^2}{\left[Zn^{2+}\right]}\left[n = 2e^-\right]$$

$$0.199V - 0.76V = \frac{0.0591}{2}\log\frac{\left[H^+\right]^2}{[0.1]}$$

$H^+ = 10^{-10}M$

$\therefore$ pH= $-\log H^+ = -\log 10^{-10}$

$pH = 10\log 10$

$pH = 10$

32. D

Sol.

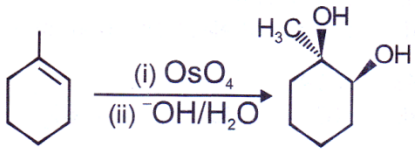
NO	$+ NO_3$	$2NO_2$
$1-\frac{x}{2}$	$3-\frac{x}{2}$	x
$3-x$	$3-x$	2x

$$K = \frac{x^2}{\left(1-\frac{x}{2}\right)\left(3-\frac{x}{2}\right)} = K = \frac{4x^2}{(3-x)^2}$$

Equating the two gives x = 3/2 then K = 4

33. C

Sol.



34. D

Sol. $C_2O_4^{2-} > CO > CO^+$

Bond order of C–O in $C_2O_4^{-2}$ is $\Rightarrow 1.5$

Bond order of C–O in CO is $\Rightarrow$ 3

Bond order of C–O in CO^+ is $\Rightarrow$ more than 3

35. D

Sol. Given $m_Y = 2m_Z$ . . . (i)

U_{rms} of 'Y' = 2 × U_{rms} of Z . . . (ii)

and no. of molecules of Y = No. of molecules of 'Z' . . . (iii)

$\because\ P_YV_Y = \frac{1}{3}M_YU^2_{rms_Y}$

$P_ZV_Z = \frac{1}{3}M_ZU^2_{rms_Z}$

$\therefore \quad \frac{P_Y V_Y}{P_Z V_Z} = \frac{M_Y}{M_Z} \times \frac{U_Y^2}{U_Z^2}$ . . . . . (iv)

$\therefore$ Given $V_Y = V_Z$ . . . . . (v)

By equation 1, 2, 3, 4, 5,

$$\frac{P_Y}{P_Z} = 2 \times (2)^2 = 8$$

36. B

Sol. Mercuric chloride, $HgCl_2$.

Chromyl chloride test is used for the conformation of chloride ion. Formation of red vapour, yellow solution, yellow ppt.

[Chloride of $Sb^{+3}, Sn^{+4}, Ag^{+1}, Pb^{+2}, Hg^{+2}, Hg^{+}, Sb^{+5}$ does not give dissolve significantly in water]

37. B

Sol. Mannose

Mannose is a monosaccharide, a C–2 epimer of glucose.

38. A

Sol. 2.5 g/g adsorbent

$\frac{x}{m} = k.P^{1/n}$ since, log k = 0.699,

Hence , k = 5;

Slope = $\frac{1}{n} = \tan 45° = 1$

Thus, $\frac{x}{m} = 5 \times 0.5 g/g$ adsorbent

39. B

Sol. Analgesic

40. A

Sol. Phenol is also called carbolic acid, Benzoic acid is more acidic than phenol. So the acid reacts with $NaHCO_3$ to give CO_2(g), while phenol does not react.

(B) Only carbonyl compound gives 2, 4-DNP test.

(C) Both acid and phenol react with aqueous NaOH to give their corresponding salts.

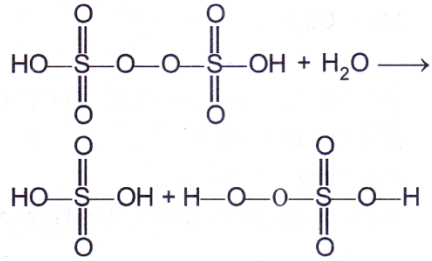
$PhCOOH + NaOH \rightarrow PhCOONa$

$PhOH + NaOH \rightarrow PhONa$

(D) Both acid and phenol react with NH_3. $PhCOOH + NH_3 \rightarrow PhNH_2$ (Aniline) + H_2O

41. C

Sol. one mole of sulphuric acid, one mole of peroxomono-sulphuric acid



42. D

Sol. The electronic configurations of Ni, Fe, CO and Cr in the given complexes are $(3d)^8$, $(3d)^6$,$(3d)^6$ and $(3d)^4$, respectively. These involves dsp^2, d^2sp^3, d^2sp^3 and sp^3 d^2-hybridizatin respectively. The complex $\left[Cr(H_2O)_6\right]SO_4$ will be paramagnetic as Cr contains four unpaired electrons.

43. A

Sol. Catalytic oxidation of NH_3 by O_2.

44. B

Sol. For gas A the temperature remains constant but in case of gas B the temperature increases so the pressure increases. Therefore, the final pressure of 'A' will be less than that of B.

45. D

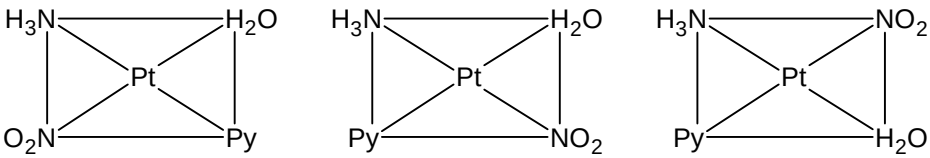
Sol. H_2O_2 is thermally unstable and it decomposes easily.

$H_2O_2(l) \rightarrow H_2O(l) + \frac{1}{2}O_2(g)$ Its decomposition is catalysed by alkali metals present in traces in the glass of the vessel.

SECTION – B

46. 3

Sol.



47. 4

Sol. Borax molecule exists structurally as

$Na_2[B_4O_5(OH)_4].\ 8H_2O$

x = 4

SECTION – C

48. 00083.50

Sol. $2C_2H_6(g) + 7O_2(g) \longrightarrow 4CO_2(g) + 6H_2O(\ell)\ \ \Delta H = -3129KJ$

$\Delta H = \sum \Delta H_{f(Product)} - \sum \Delta H_{f(reactants)}$

$-3129 = \left[4 \times (-395) + 6 \times -(286)\right] - \left[2 \times \Delta H_f C_2H_6 + 7 \times 0\right]$

$\Delta H_{fC_2H_6} = -83.5KJ$

49. 00010.60

Sol. $\Lambda_\infty NH_4Cl = \lambda NH_4^+ + \lambda Cl^- = 130$ … (1)

$\Lambda_\infty NaOH = \lambda Na^+ + \lambda HO^- = 220$ … (2)

$\Lambda_\infty NaCl = \lambda Na^+ + \lambda Cl^- = 100$ … (3)

[(1) + (2)] – [(3)]

$$\lambda_\infty NH_4^+ + \lambda_\infty HO^- = 250 = \Lambda_\infty NH_4OH$$

$$'\alpha' \text{ of } NH_4OH = \frac{\Lambda_m^c}{\Lambda_\infty} = \frac{10}{250} = \frac{1}{25}$$

$$\left[HO^-\right] = C\alpha = \frac{1}{100} \times \frac{1}{25} \Rightarrow 4 \times 10^{-4}$$

$$pH = 10 + 2\log 2 = 10.6$$

50. 00002.69

Sol. $HA \rightleftharpoons H^+ + A^-$

$[H^+] = C\alpha = 10^{-2} \quad \therefore \ \alpha = 0.1$

$i = 1 + \alpha = 1.1 \ \Rightarrow \ \pi = i(CRT)$

Thus $\pi = 0.1 \times 0.0821 \times 298 \times 1.1 = 2.69$ atm

Mathematics

PART – III

SECTION – A

51. A

Sol. Directrix: $x-3=-\frac{1}{2}$, $x=3-\frac{1}{2}=\frac{5}{2}$ ..... (1)

Slope of tangent at P(5, 2): $2y\frac{dy}{dx}=2$, $\frac{dy}{dx}=\frac{1}{2}$

Equation of tangent $y-2=\frac{1}{2}(x-5)$ ..... (2)

$y-2=\frac{1}{2}\left(\frac{5}{2}-5\right)=-\frac{5}{4}$, $y=2-\frac{5}{4}=\frac{3}{4}$

$Q=\left(\frac{5}{2},\frac{3}{4}\right)$ circumcentre is mid-point of P and $Q\equiv\left(\frac{5+\frac{5}{2}}{2},\frac{2+\frac{3}{4}}{2}\right)=\left(\frac{15}{4},\frac{11}{8}\right)$

52. C

Sol. $\int_0^2 f(x)dx=\int_0^2 f(x)dx+\int_0^2 f(x)dx=\int_0^1 (f(x)+f(x+1))dx$

$=\int_0^1 (x^2+2x+3)dx=\frac{x^3}{3}+x^2+3x\Big|_0^1=\frac{1}{3}+1+3=\frac{13}{3}$

53. D

Sol. Case 1: $1\times4\times5\times5=100$

Case 2: $2\times5\times5\times5=\frac{250}{350}$

One number will be 2100

So required number of ways = 349

54. D

Sol. $\lim\limits_{x\to\infty}\frac{\frac{30}{x^{\frac{1}{2}}}+3+x^{-\frac{1}{6}}}{\frac{3}{\sqrt{x}}+\sqrt{9-\frac{5}{x}}+x^{-\frac{1}{6}}\left(27-\frac{4}{x}\right)^{\frac{1}{3}}}=\frac{0+3+0}{0+3+0}=1$

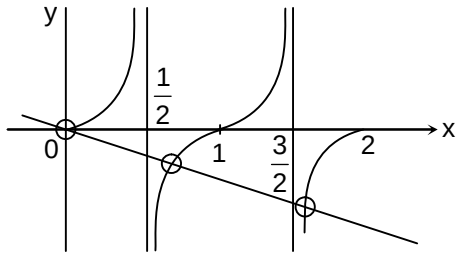
55. B

Sol. $f'(x)=\sin\pi x+\pi x\cos\pi x=0$

$\sin\pi x=-\pi x\cos\pi x$

$\tan\pi x=-\pi x$

3 solutions



56. B

Sol. $\cos^{-1} x + \sin^{-1} x + \tan^{-1} x = \pi - \cot^{-1} x$ for $x \in [0, 1]$

$\frac{\pi}{2} + \tan^{-1} x = \pi - \cot^{-1} x$, $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$

For $x \in [-1, 0)$, $\cos^{-1} x - \sin^{-1} x - \tan^{-1} x = \pi - \cos^{-1} x$ does not satisfy

So, $[a, b] = [0, 1]$, $a + b = 1$

57. C

Sol. $\frac{1-\cos 4x}{2} = \frac{1+\cos 2x}{2}$, $\cos 4x + \cos 2x = 0$, $2 \cos 3x \cos x = 0$

$x = (2n+1)\frac{\pi}{6}$, $\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{9\pi}{6}, \frac{11\pi}{6}$

58. C

Sol. Equation of AB as chord of contact $T = 0$

$x\alpha + y\beta - 16 = 0$ ..... (1)

Also, equation of AB $2x + y = 3$ ..... (2)

$\frac{\alpha}{2} = \frac{\beta}{1} = \frac{16}{3}$, $\alpha = \frac{32}{3}$, $\beta = \frac{16}{3}$, $\alpha + \beta = \frac{48}{3} = 16$

59. C

Sol. Let α, β, γ are roots then $f(x) = (x - \alpha)(x - \beta)(x - \gamma)$, $(5 - \alpha)(5 - \beta)(5 - \gamma) = 2$

$5 - \alpha = 1$, $5 - \beta = 1$, $5 - \gamma = 2$, $\alpha = 4$, $\beta = 4$, $\gamma = 3$, $a = 4 + 4 + 3 = 11$

60. C

Sol. $3x^2 + 4y^2 = 1$, Tangent at $P(x_1, y_1)$

$3xx_1 + 4yy_1 = 1$ ..... (1)

$x + y = 1$ ..... (2)

Comparing $\frac{3x_1}{1} = \frac{4y_1}{1} = 1$, $x_1 = \frac{1}{3}$, $y_1 = \frac{1}{4}$, $3x_1 + 8y_1 = 3$

61. C

Sol. $[x] + [2x] = 4$

Case-I: $[x] = 0$, $[2x] = 4$ not possible

Case-II: $[x] = 1$, $[2x] = 3$

$1 \le x < 2$ $\quad 3 \le 2x < 4$

$\frac{3}{2} \le x < 2$ possible

$[x] = 2$ $\quad [2x] = 2$ not possible

62. C

Sol. $f(x) = \frac{e^x - e^x}{10e^x + e^x} = 0$ for $x \ge 0$

$= \frac{e^x - e^{-x}}{10e^x + e^{-x}} = \frac{e^{2x} - 1}{10e^{2x} + 1}$ for $x < 0$

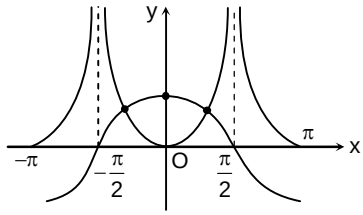
For $x < 0$, $f(x) = \frac{2e^{2x} - 1}{10\left(e^{2x} + \frac{1}{10}\right)} = \frac{e^{2x} + \frac{1}{10} - \frac{11}{10}}{10\left(e^{2x} + \frac{1}{10}\right)} = \frac{1}{10}\left[1 - \frac{\frac{11}{10}}{e^{2x} + \frac{1}{10}}\right]$

$\frac{1}{10} < e^{2x} + \frac{1}{10} < \frac{1}{10} + 1$, $f(x) \in (-1, 0)$. Range of $f(x)$ is $(-1, 0]$

63. A

Sol. $f(x) = \max(|\tan x|, \cos|x|)$ is $(-\pi, \pi)$

3 critical points



64. C

Sol. Area of ΔCQR = constant = ab = $2\sqrt{3}$

65. B

Sol. $(a + bw + cw^2) = (a + bw + cw^2)(a + bw^2 + cw) = a^2 + b^2 + c^2 - ab - bc - ac$

$= \frac{(a-b)^2 + (b-c)^2 + (c-a)^2}{2} = \frac{1^2 + 2^2 + 3^2}{2} = 7$

66. B

Sol. Let $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4} = \lambda$, $x = 2\lambda + 1$, $y = 3\lambda - 1$, $z = 4\lambda + 1$ lies on 2nd line $\frac{x-3}{1} = \frac{y-k}{2} = \frac{z}{1}$

$\frac{2\lambda+1-3}{1} = \frac{3\lambda-1-k}{2} = \frac{4\lambda+1}{1}$. So, $2\lambda - 2 = 4\lambda + 1$, $2\lambda = -3 \Rightarrow \lambda = -\frac{3}{2}$

$\alpha = -2$, $\beta = -\frac{11}{2}$, $\gamma = -5$, $\alpha + \gamma - 2\beta = -2 - 5 + 11 = 4$

67. A

Sol. Shown in the table

p	q	$\sim(p \vee q)$	$\sim p \wedge q$	$\sim(p \vee q) \vee (\sim p \wedge q)$
T	T	F	F	F
T	F	F	F	F
F	T	F	T	T
F	F	T	F	T

68. B

Sol. Sum 4 = {(1, 3), (2, 2), (3, 1)}

Sum 7 = {(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)}

$n(E) = 6$, $n(S) = 9$, $p(E) = \frac{6}{9} = \frac{2}{3}$

69. B

Sol. $\alpha = \vec{a} + \vec{a}\times\vec{b}\cdot\left(\vec{b} + \left(\vec{a}\times\vec{b}\right)\right)\times\left(\vec{a}\times\vec{b}\right) = \left(\vec{a} + \vec{a}\times\vec{b}\right)\cdot\left(\left(\vec{b}\cdot\vec{b}\right)\vec{a} - \left(\vec{b}\cdot\vec{a}\right)\vec{b}\right) = 1 - 0 + 0 + 0 = 1$

$\beta = \begin{vmatrix} \vec{a}\cdot\vec{b} & \vec{a}\cdot i \\ \vec{b}\cdot\vec{c} & i\cdot\vec{i} \end{vmatrix} + \begin{vmatrix} \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{j} \\ \vec{b}\cdot\vec{j} & \vec{j}\cdot\vec{j} \end{vmatrix} + \begin{vmatrix} \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{k} \\ \vec{b}\cdot\vec{k} & \vec{k}\cdot\vec{k} \end{vmatrix} = 2\left(\vec{a}\cdot\vec{b}\right) = 0$

70. B

Sol. $z^2 = \bar{z}$, $\Rightarrow \bar{z}^2 = z \Rightarrow z^4 = z \Rightarrow$ 4 solutions

So, coefficient of x^{11} in $(2x^2 + x - 3)^6$

= coefficient of x^{11} in $(x - 1)^6(2x + 3)^6 = {}^6C_0 \times {}^6C_1 2^5 \times 3 - {}^6C_1 \times {}^6C_0 2^6$

$= 3 \times 6 \times 2^5 - 6 \times 2^6 = 6 \times 2^5 (3 - 2) = 192$

SECTION – B

71. 2

Sol. $\frac{dy}{dx}=\frac{y(1+xy)}{x}=\frac{y}{x}+y^2 \Rightarrow \frac{1}{y^2}\frac{dy}{dx}-\frac{1}{xy}=1$

Put $\frac{1}{y}=t$, $-\frac{1}{y^2}\frac{dy}{dx}=\frac{dt}{dx}$, $-\frac{dt}{dx}-\frac{1}{x}t=1$, $\frac{dt}{dx}+\frac{t}{x}=-1$

If = $e^{\int\frac{1}{x}dx}=x$, $t\cdot x=\int -1\cdot x\,dx$, $\frac{x}{y}=-\frac{x^2}{2}+c$, $\frac{4}{2}=-8+c \Rightarrow c=10$

$\frac{x}{y}=10-\frac{x^2}{2} \Rightarrow y=\frac{2x}{20-x^2}$, $f(4)=\frac{8}{20-16}=2$

72. 4

Sol. $5\sin x=\ell(\sin x-2\cos x)+m(\cos x+2\sin x)$, $\ell+2m=5$, $m-2\ell=0 \therefore \ell=1, m=2$

So, $\int\frac{5\sin x}{\sin x-2\cos x}dx=\int 1\cdot dx+2\int\frac{\cos x+2\sin x}{\sin x-2\cos x}dx = x+2\ln|\sin x-2\cos x|+c$

A = 1, B = 2, 2A + B = 4

SECTION – C

73. 00040.50

Sol. $\Sigma x=210$, $\Sigma x^2=2530$

Increase in $\Sigma x=10$ then $\Sigma x'=220$

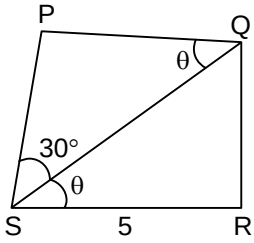
Increase in $\Sigma x^2=1600-900=700$

$\Sigma x'^2=2530+700=3230$

Variance = $\frac{1}{n}\Sigma x'^2-\left(\frac{\Sigma x'}{n}\right)^2=\frac{1}{20}\times 3230-\left(\frac{220}{20}\right)^2=\frac{323}{2}-121=\frac{323-242}{2}=\frac{81}{2}=40.50$

74. 00003.71

Sol. $SQ=\sqrt{25+27}=\sqrt{52}$ and let $\angle QSR=\theta$, then $\angle PQS=\theta$



$\angle SPQ=180°-30°-\theta=150°-\theta=\pi-\theta-30°$

In ΔPQS, $\frac{SQ}{\sin(\pi-\theta-30°)}=\frac{PQ}{\sin 30°}$, $\frac{\sqrt{52}}{\sin(\theta+30°)}=\frac{PQ}{\frac{1}{2}}$

$PQ=\frac{\sqrt{13}}{\sin\theta\cos 30°+\sin 30°\cos\theta}$

$=\frac{\sqrt{13}}{\frac{3\sqrt{3}}{\sqrt{52}}\times\frac{\sqrt{3}}{2}+\frac{1}{2}\times\frac{5}{\sqrt{52}}}=\frac{\sqrt{13}\times 2\sqrt{52}}{9+5}$

$=\frac{52}{14}=\frac{26}{7}=3.714$

75. 00004.33

Sol. $x=\frac{1}{1-\frac{1}{4}}=\frac{4}{3}$, $y=\frac{1}{1-\frac{1}{2}}=2$, $z=\frac{1}{1-\frac{1}{c}}$, $\frac{2}{y}=\frac{1}{x}+\frac{1}{z}$, $1=\frac{3}{4}+1-\frac{1}{c}$, $c=\frac{4}{3}$